

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 4A

Organising and summarising data

Bellerbys
Preparation for University Education

EXERCISE A0

Week 3 revision

Slides 3-4

1. For 2, 4, 4, 6, 9, calculate the mean and median.

Mean: _____ Median: _____

2. Which has more spread: 4, 6, 8 or 2, 6, 10? Give a reason.

3. A: mean 20, SD 4. B: mean 40, SD 6. Which has less relative variation?

$CV = \frac{SD}{mean} \times 100\%$ A: _____ B: _____

Choice and reason: _____

EXERCISE A1a

Count the lunch choices

Slides 6-7

40 students each chose one lunch. Tally every response once.

S S S D D R N S D D
S N N R N R D D N D
N R D D S N R S N N
R R S N R N D R N R

N = noodles R = rice bowls S = sandwiches D = dumplings

Lunch option	Tally	Frequency
Noodles		
Rice bowls		
Sandwiches		
Dumplings		
Total		

EXERCISE A1b

Find the shares

Slides 9-10

Use your frequencies above. Relative frequency = $f \div n$; $n = 40$.

Lunch option	Relative frequency	Percentage
Noodles		
Rice bowls		
Sandwiches		
Dumplings		
Total		

What percentage chose noodles OR rice bowls? _____

SHOWING CATEGORIES

EXERCISE A2 Draw a drink-orders bar chart

Slides 13-14

The café recorded 50 orders. Choose an even scale starting at zero.
Label both axes, draw bars with gaps, and add a title.

Drink	Orders
Milk tea	18
Fruit tea	14
Juice	10
Water	8



Chart title: _____

EXERCISE A3a Calculate the sector angles

Slide 17

Use the lunch frequencies from A1a. The first sector is the worked example.

Lunch option	Frequency	Angle
Noodles	12	108°
Rice bowls		
Sandwiches		
Dumplings		
Total		

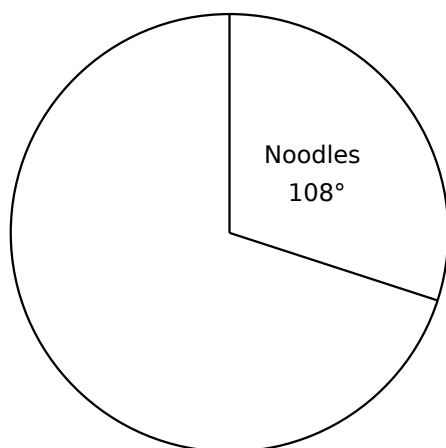
$$\text{Angle} = \frac{f}{n} \times 360^\circ$$

f = category frequency
n = total responses (40)

Check: angles total 360°.

EXERCISE A3b Complete the pie chart

Slides 19-20



Keep the noodles sector.
Measure from the previous radius.
Label each category and percentage.

A3: Which chart makes the difference between noodles and sandwiches clearer? Explain.

GROUPING NUMERICAL DATA

EXERCISE A4a Place each value in a class

Slide 24

Mixed values: 34, 19, 42, 20, 29, 11, 30, 48

$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$

Explain why 19 and 20 belong to different classes:

EXERCISE A4b Build the grouped table

Slides 28-29

A campus food club recorded sign-ups at 25 promotion sessions.

Each session lasted 30 minutes. Each number is the count from one session.

35 22 48 31 15
 40 28 34 43 19
 36 23 39 21 44
 27 32 30 42 25
 29 33 41 38 37

Tally, find the midpoint, and calculate %.

Use $\text{lower} \leq x < \text{upper}$ for every class.
$$\text{Midpoint} = \frac{\text{lower boundary} + \text{upper boundary}}{2}$$

Class	Tally	f	Midpoint	%
$10 \leq x < 20$				
$20 \leq x < 30$				
$30 \leq x < 40$				
$40 \leq x < 50$				
Total				

Check: frequencies total 25; percentages total 100%. Keep this table for Class B.

EXERCISE A5 Estimate sign-ups per session

Slides 34-37

Copy f and m from A4b. Complete $f \times m$, then calculate the estimated mean.

Sign-ups	f	m	$f \times m$
$10 \leq x < 20$			
$20 \leq x < 30$			
$30 \leq x < 40$			
$40 \leq x < 50$			
Total			

Estimated mean $\approx \frac{\sum(fm)}{n} =$ sign-ups per session

KEEPING THE ORIGINAL VALUES

EXERCISE A6

Ten delivery times

Slides 41-43

Delivery times (minutes): 27, 32, 24, 41, 27, 21, 38, 30, 26, 24

Put the stems and leaves in order. Keep repeats and include a key.

Stem	Leaf

Key: _____

Check: number of leaves = _____

Write all ten values in order:

--	--	--	--	--	--	--	--	--	--

Middle positions: _____ Middle values: _____

Calculate the median. Show your working and include the unit.

Slide 44 check: if 2 | 8 means 2.8 hours, what does 11 | 2 mean?

EXERCISE A7

Three quick checks

Slides 45-46

1. What percentage is 4 out of 25?

2. What is the midpoint of $40 \leq x < 50$?

3. When would we estimate a mean using midpoints?
